

AUTOSAR High-Level Design - Version 1

1. System Overview

Project: Vehicle Control Platform

Architecture Type: AUTOSAR Classic

Purpose: Coordinate vehicle control and sensor data.

2. Software Components

Software Component: EngineControl

Software Component: SensorManager

Software Component: CommunicationManager

3. Interfaces

Interface: EngineStatus

Interface: SensorData

Interface: VehicleCommunication

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4. Ports

Port: EngineControl_StatusPort

Port: SensorManager_DataPort

Port: CommunicationManager_NetworkPort

5. Signals

Signal: EngineSpeed

Signal: VehicleSpeed

Signal: EngineTemperature

6. Dependencies

Dependency: EngineControl -> SensorManager

Dependency: SensorManager -> CommunicationManager

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7. Functional Flow

SensorManager acquires vehicle sensor information.

EngineControl consumes sensor information.

CommunicationManager handles external communication.

8. Architecture Notes

EngineControl uses EngineStatus for engine state information.

SensorManager provides sensor-related data to consumers.

CommunicationManager provides network communication services.

9. Review Notes

Architecture shall be reviewed before implementation.

Interfaces and dependencies shall remain traceable.